Multiclass Brain Tumor Classification Using Convolutional Neural Network and Support Vector Machine.

2021 Mohammad Ali Jinnah University International Conference on Computing (MAJICC) | 978-1-6654-2413-4/21/$31.00 ©2021 IEEE | DOI: 10.1109/MAJICC53071.2021.9526262

Hareem Kibriya[1], Momina Masood[2], Marriam Nawaz, Rimsha Rafique, Safia Rehman
Department of Computer Sciences
University of Engineering and Technology
Taxila, Pakistan
hareem.kibriya@students.uettaxila.edu.pk[1], momina.masood@uettaxila.edu.pk[2]

Abstract— **Brain tumor is caused by abnormal growth of cells inside the skull. It is a lethal disease, the diagnosis of which is a difficult task for radiologists. Most of the tumors are misdiagnosed due to the variability and complexity of lesions, which reduces the survival rate in patients. Diagnosis of brain tumors via computer vision algorithms is a challenging task. Traditional brain tumor identification techniques require manual segmentation or handcrafted feature extraction that is error-prone and time-consuming. This research explores multiclass brain tumor classification methods using Deep Learning (DL) and Machine Learning (ML) techniques. First, the brain MRI images are classified via end-to-end Convolutional Neural Network (CNN) models i.e. ResNet-18 and GoogLeNet. The deep features extracted from the CNN models are also classified using Support Vector Machine (SVM). The proposed method is trained and evaluated on 15,320 MRI images and achieved the highest accuracy of 98% via CNN-SVM based method. Our proposed method outperformed the existing brain tumor identification systems and can assist the doctors to detect brain tumors and make key decisions related to the patient's treatment.**

Keywords—**Deep Learning, Computer Vision, Brain Tumor**

I. INTRODUCTION

Brain is a vital organ that is responsible for controlling all voluntary and involuntary processes in the human body. The brain can be affected by many diseases including brain tumors that are usually caused by uncontrolled growth of cells inside the brain [1] These tumors are very hard to treat [2], so early diagnosis plays a vital role in the patient's health. WHO predicted that brain tumors will be increased by 5% every year all around the world [3].

The interpretation of tumor type by professionals before surgery is a complicated task because the lesions vary in intensity and shape. Misdiagnosis of tumor type can decrease the chance of a patient's survival. Traditional brain tumor detection approaches require manual lesion segmentation and handcrafted feature extraction before classification that is time-consuming and suffers from inter-observer variability. On the other hand, the CNNs can automatically extract deep features and perform classification [4, 5].

In recent years, DL and ML are dominating medical image diagnosis problems. DL and ML based medical image classification algorithms provide an opinion to medical professionals in making therapy-related decisions. CNNs are widely being applied to image analysis tasks i.e. deep feature extraction, segmentation and classification. An automated brain tumor classification system can help physicians decide upon successful treatment options [6].

Many researchers have developed automated brain tumor identification techniques. A detailed summary of some of the existing brain tumor classification techniques is provided in TABLE 1. The table discusses the DL techniques employed, the dataset used to conduct those experiments and the accuracy percentage obtained. The authors in [4, 7] presented a VGG based brain tumor classification system. [3, 8] proposed a Residual Network-based technique whereas, [4, 9] presented AlexNet based approach.

This paper presents an automated approach employing DL and ML algorithms to classify brain tumors in MRI images. The proposed method is evaluated on a publically available dataset [10] and can assist doctors in making decisions regarding the therapy of the patients. The remaining paper is arranged as follows: Section 2 discusses the proposed methodology. Section 3 describes the experimental setup. Section 4 presents the results. Finally, the conclusion is presented in Section 5.

TABLE 1. SUMMARY OF EXISTING BRAIN TUMOR IDENTIFICATION SYSTEMS

Reference	Year	DL Technique	Dataset	ACC %
Cinar et al. [3]	2020	ResNet-50	Kaggle	97
Swati et al. [4]	2019	VGG (16,19) , AlexNet	Figshare	94
Anarkari et al. [5]	2019	Custom CNN	Figshare	94
Naser et al. [7]	2020	U-Net, VGG-16	TCIA	92
Sharma et al. [8]	2020	ResNet	SICAS Local	93
Hashemzehi et al. [9]	2020	AlexNet, ConvNADE	Figshare	96

978-1-6654-2413-4/21/$31.00 ©2021 IEEE

Authorized licensed use limited to: Cambridge Institute of Technology - Bengaluru. Downloaded on March 04,2026 at 16:14:01 UTC from IEEE Xplore. Restrictions apply.

II. PROPOSED METHODOLOGY

In the proposed method, we employed DL and ML-based algorithms to identify and classify Pituitary, Meningioma and Glioma tumors in MRI images. The elaborate diagram of the proposed scheme is shown in Fig 1. Before feeding the images to the CNNs for classification, they are preprocessed and artificially augmented.

A. Image Preprocessing

During this phase, the images are normalized using the min-max normalization technique as discussed in equation (1).

$$f(x,y) = \frac{f(x,y) - Z_{min}}{Z_{max} - Z_{min}} \quad (1)$$

The brain image is denoted by f, whereas, Z_{max} and Z_{min} represent the maximum and minimum pixel values in the image. After the normalization process, the images are resized [224 224 3] to match the size of the input layer of the CNN architectures.

B. Data Augmentation

In the next step, we increased the data samples by artificially augmenting the data. Data augmentation reduces the problems of imbalanced classes so that the model is not biased towards classifying instances as majority class types [11]. We augmented images by adding salt and pepper noise, left/right mirroring, flipping around the x-axis and rotating the images to 45 degrees. These augmentation techniques increased the dataset by a factor of 5 from 3064 images to 15,320 images. TABLE II shows the number of images before and after augmentation. Fig 2. displays different image augmentation techniques applied to the MRI dataset.

C. Convolutional Neural Network

CNN is an efficient supervised learning method that has made remarkable improvements in the field of computer vision. In our proposed method we applied two deep transfer learned models i.e. GoogLeNet [12] ResNet-18 [13] to a three-class brain tumor classification problem. Transfer learning is a commonly used technique in DL that aims to improve learning in the target domain using the source domain and the learning task [14].

Training the DL models from scratch is computationally intensive as it requires a lot of resources, so the transfer learning frameworks aim to solve the resource problem [13, 15].

In this study, we fine-tuned the CNN models by changing the last three layers of the architecture according to our target domain. The Fully Connected (FC) layer from the original architecture was replaced by a new one. Similarly, we also replaced the Softmax layer and the classification layer from both architectures. Fig 3. presents the modifications made in the CNN architectures for this study.

We also classified the deep CNN features using an SVM classifier with an error-correcting output code (ECOC) model. SVM belongs to category of supervised learning algorithms. It has gained popularity due to promising results and is employed in various machine learning researches all over the world. It is memory efficient and effective in high-dimensional spaces [16].

III. EXPERIMENTAL SETUP

A. Dataset

To conduct this study, we used a publically available Figshare dataset that contains 3064 brain MRI images of 233 patients. It is composed of 708 samples of Meningioma, 930 samples of Pituitary and 1426 samples of Glioma. The final version of the dataset was uploaded in 2017 [10].

B. Implementation Details

For implementation, we used Matlab 2019a on Intel Core i5 6300 CPU with 8 GB RAM. The dataset was split randomly into a 70% training set for training the network and a 30% validation set to evaluate the trained model impartially.

TABLE II. NUMBER OF MRI SAMPLES BEFORE AND AFTER AUGMENTATION

Tumor Type	**MRI samples before Augmentation**	**MRI samples after Augmentation**
Glioma	708	3540
Meningioma	1426	7130
Pituitary	930	4650
Total	**3064**	**15,320**

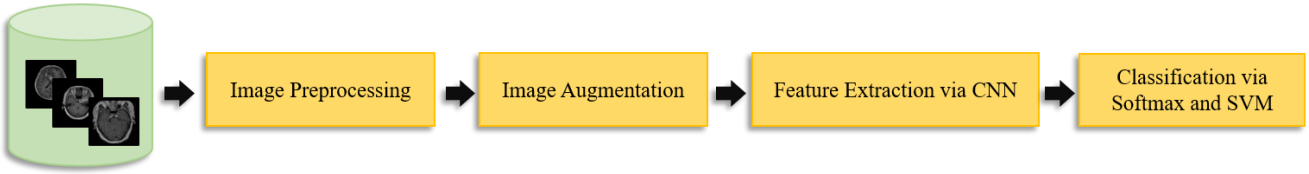
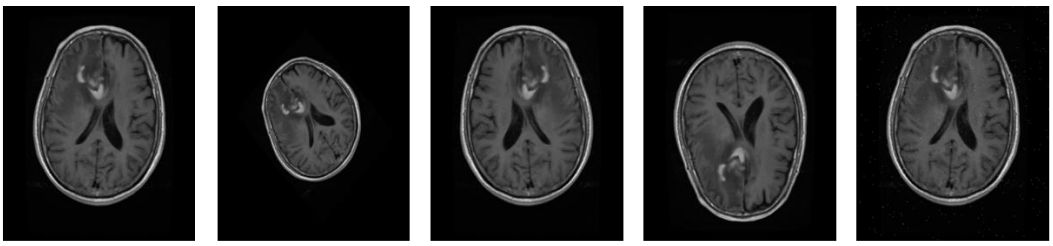


Fig 1. Proposed Method Design

Authorized licensed use limited to: Cambridge Institute of Technology - Bengaluru. Downloaded on March 04,2026 at 16:14:01 UTC from IEEE Xplore. Restrictions apply.



(a) Original Image (b) 45 degree rotation (c) Right/Left mirroring (d) Upside down flip (e) Salt/Pepper Noise

Fig 2. Image Augmentation Techniques

IV. RESULTS

A. Evaluation Parameters

The proposed method is evaluated via Precision (PRE), Recall (REC) and Accuracy (ACC). PRE is mentioned in Equation (2), REC in Equation (3), and ACC in Equation (4).

$$PRE = \frac{tp}{tp + fp} \quad (2)$$

$$REC = \frac{tp}{tp + fn} \quad (3)$$

$$ACC = \frac{tp + tn}{tp + tn + fp + fn} \quad (4)$$

where tp denotes true positives, tn represents true negatives, fp denotes false positives and fn represents false negatives.

B. Proposed method results

In this study, we trained and evaluated two end-to-end DL models to identify brain tumors in MRI images. We then supplied the deep feature vectors as input to SVM with an ECOC model for final output. End-to-end learning models i.e. GoogLeNet and ResNet-18 achieved an accuracy of 97.4% and 97.8% respectively. The detailed results of CNNs in terms of ACC, PRE and REC are presented in Fig 4. The graph indicates that ResNet-18 performed better than GoogLeNet to classify brain tumors.

SVM classified deep feature vector from ResNet-18 and GoogLeNet with an accuracy of 98% and 97.6% respectively It is observed that the proposed system's accuracy increased when SVM was used instead of the Softmax layer of the CNN. TABLE III discusses the classification results obtained via SVM in terms of ACC, PRE and REC.

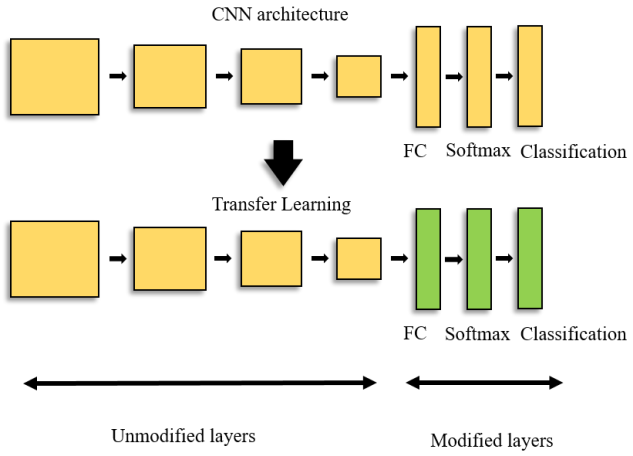


Fig 3. Modifying CNN for Application

C. Comparison with state-of-the-art

TABLE IV presents an elaborate result comparison of the proposed method with the existing brain tumor classification techniques. Our proposed method succeeded in achieving an accuracy of 97.8% via ResNet-18, and 97.4% via GoogLeNet. Whereas SVM obtained 98% accuracy to classify deep feature vector from ResNet-18 and 97.6% accuracy to classify feature vector from GoogLeNet. The results are proof that the proposed method provides better performance than the existing techniques.

TABLE III. RESULTS OBTAINED FROM SVM

Deep feature vector	**ACC %**	**PRE %**	**REC %**
GoogLeNet	97.6	97.3	97.3
ResNet-18	98.0	98.3	98.0

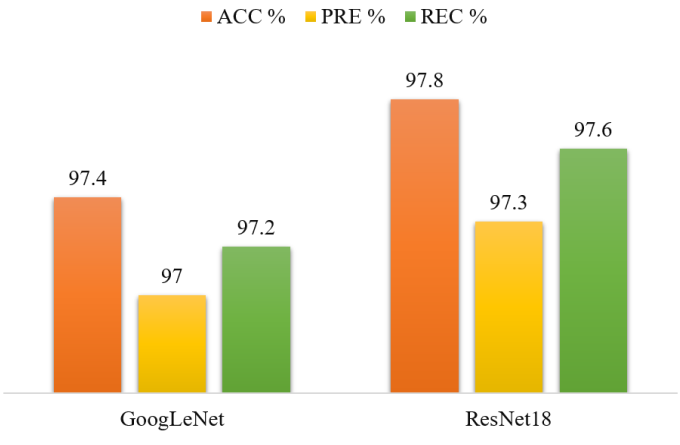


Fig 4. Results Obtained via End-to-End Classification Models

Authorized licensed use limited to: Cambridge Institute of Technology - Bengaluru. Downloaded on March 04,2026 at 16:14:01 UTC from IEEE Xplore. Restrictions apply.

TABLE IV. COMPARISON OF PROPOSED METHOD WITH EXISTING TECHNIQUES

Ref.	Year	DL Technique	ACC %
Swati et al. [4]	2019	AlexNet	89.9
Hashemzehi et al. [9]	2020	Hybrid CNN-NADE	95.0
Cinar et al. [3]	2020	ResNet-50	97.2
Sharma et al. [8]	2020	ResNet	93.0
Pundir et al. [14]	2021	VGG-16	91.8
Sejuti et al. [17]	2021	CNN+SVM	97.1
Noreen et al. [18]	2021	Inception-v3-Ensemble	94.3
Proposed Method	**2021**	**ResNet-18**	**97.8**
		GoogLeNet	**97.4**
		ResNet-18 + SVM	**98.0**
		GoogLeNet + SVM	**97.6**

V. CONCLUSION

This paper introduces an automated brain tumor classification method using DL and ML based techniques. The MRI images are preprocessed, augmented and supplied to fine-tuned CNN models i.e. ResNet-18 and GoogLeNet for tumor classification. The deep feature vectors are also supplied to SVM classifier for final output. The proposed method is evaluated using ACC, SEN, and SPE. The highest performing CNN architecture i.e. ResNet-18 achieved 97.8% accuracy; whereas, the combination of ResNet-18 and SVM obtained 98% accuracy. From the results, it is evident that the proposed method is robust and efficient in classifying brain tumors from MRI images.

REFERENCES

[1] H. H. Sultan, N. M. Salem, and W. Al-Atabany, "Multi-Classification of Brain Tumor Images Using Deep Neural Network," *IEEE Access,* vol. 7, pp. 69215-69225, 2019.

[2] M. Sajjad, S. Khan, K. Muhammad, W. Wu, A. Ullah, and S. W. Baik, "Multi-grade brain tumor classification using deep CNN with extensive data augmentation," *Journal of computational science,* vol. 30, pp. 174-182, 2019.

[3] A. Çinar and M. Yıldırım, "Detection of tumors on brain MRI images using the hybrid convolutional neural network architecture," *Medical Hypotheses,* p. 109684, 2020.

[4] Z. N. K. Swati *et al.*, "Brain tumor classification for MR images using transfer learning and fine-tuning," *Computerized Medical Imaging Graphics,* vol. 75, pp. 34-46, 2019.

[5] A. K. Anaraki, M. Ayati, and F. Kazemi, "Magnetic resonance imaging-based brain tumor grades classification and grading via convolutional neural networks and genetic algorithms," *Biocybernetics Biomedical Engineering,* vol. 39, no. 1, pp. 63-74, 2019.

[6] S. Hussain, S. M. Anwar, and M. Majid, "Segmentation of glioma tumors in brain using deep convolutional neural network," *Neurocomputing,* vol. 282, pp. 248-261, 2018.

[7] M. A. Naser and M. J. Deen, "Brain tumor segmentation and grading of lower-grade glioma using deep learning in MRI images," *Computers in Biology and Medicine,* p. 103758, 2020.

[8] P. Sharma, I. Wahlang, S. Sanyal, and A. K. Maji, "Classification of Brain MRI Using Deep Learning Techniques," in *Soft Computing: Theories and Applications*: Springer, 2020, pp. 559-569.

[9] R. Hashemzehi, S. J. S. Mahdavi, M. Kheirabadi, and S. R. Kamel, "Detection of brain tumors from MRI images base on deep learning using hybrid model CNN and NADE," *Biocybernetics and Biomedical Engineering,* 2020.

[10] C. Jun, *brain tumor dataset*. 2017.

[11] C. Shorten and T. Khoshgoftaar, "A survey on image data augmentation for deep learning. J. Big Data 6 (1), 1–48 (2019)," ed.

[12] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 1-9.

[13] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770-778.

[14] A. Pundir and E. R. Kumar, "Brain Tumor Classification in MRI Images Using Transfer Learning," in *Machine Learning for Intelligent Multimedia Analytics*: Springer, 2021, pp. 307-319.

[15] H. Kibriya, R. Rafique, W. Ahmad, and S. Adnan, "Tomato Leaf Disease Detection Using Convolution Neural Network," in *2021 International Bhurban Conference on Applied Sciences and Technologies (IBCAST)*, 2021, pp. 346-351: IEEE.

[16] V. Jakkula, "Tutorial on support vector machine (svm)," *School of EECS, Washington State University,* vol. 37, 2006.

[17] Z. A. Sejuti and M. S. Islam, "An Efficient Method to Classify Brain Tumor using CNN and SVM," in *2021 2nd International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*, 2021, pp. 644-648: IEEE.

[18] N. Noreen, S. Palaniappan, A. Qayyum, I. Ahmad, and M. O. Alassafi, "Brain Tumor Classification Based on Fine-Tuned Models and the Ensemble Method," *CMC-COMPUTERS MATERIALS & CONTINUA,* vol. 67, no. 3, pp. 3967-3982, 2021.

Authorized licensed use limited to: Cambridge Institute of Technology - Bengaluru. Downloaded on March 04,2026 at 16:14:01 UTC from IEEE Xplore. Restrictions apply.